

Project Update



CP02 Group 3:

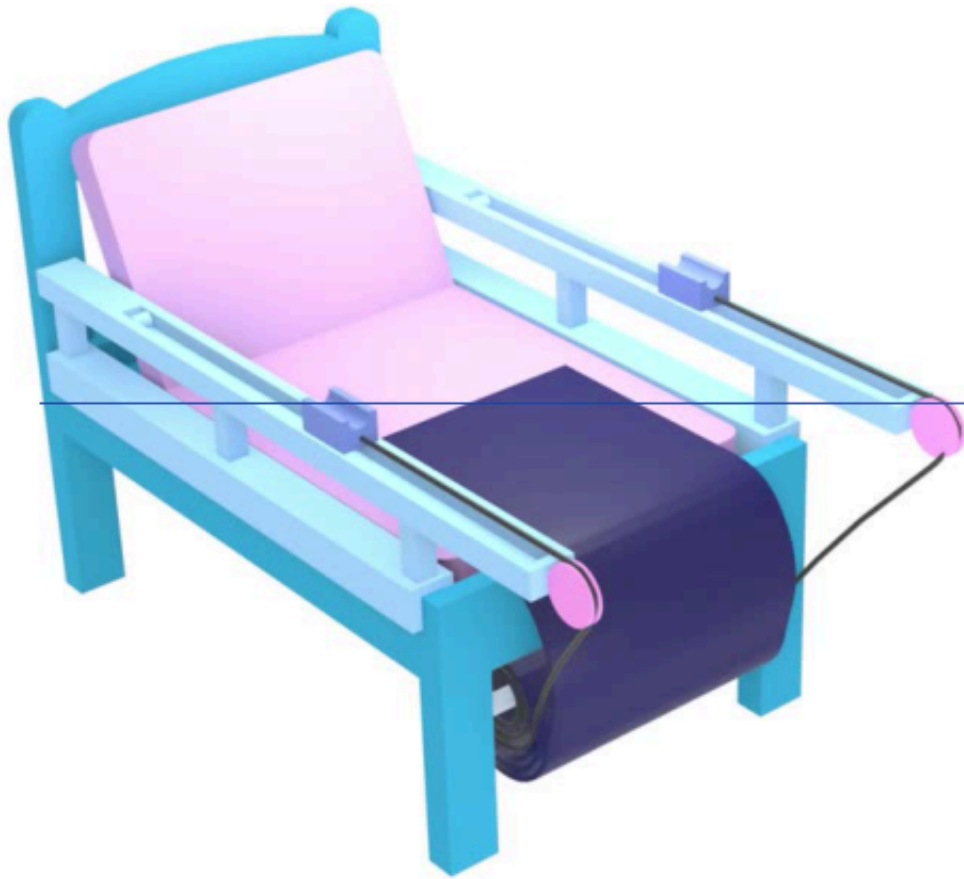
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Problem Statement

How might we enable elderly with limited strength but some level of mobility to safely transition between sitting on a bed and standing independently?



Alleviate



Affordable modular add-on to existing beds

Forward momentum to mimic natural standing motion

Enables independence and self-reliance

Reduces caregiver strain

Idea Update

Before

Pulley system syncing mat and rail movement



Uni-directional

After

**Two shafts;
Thin string connected to back of mat
to aid in pulling the mat back onto
bed**



**Bi-directional:
Can get out and get back in bed!**

Possible addition

**Lying to sitting function:
A simpler and less costly option for
people**



Our Progress



Completed

Purchase of logistics

Assembling circuit

Doing Arduino code

CADDing and printing required parts

CaDDE:
Modelled the motor speed & shaft rotation angle required for different users

Pending

Assembling Physical and Electronic Parts - pending shaft delivery

Testing

Circuits & Electronics:

Test actual circuit with Arduino Code

Assess the power consumption, safety, and tunability of performance

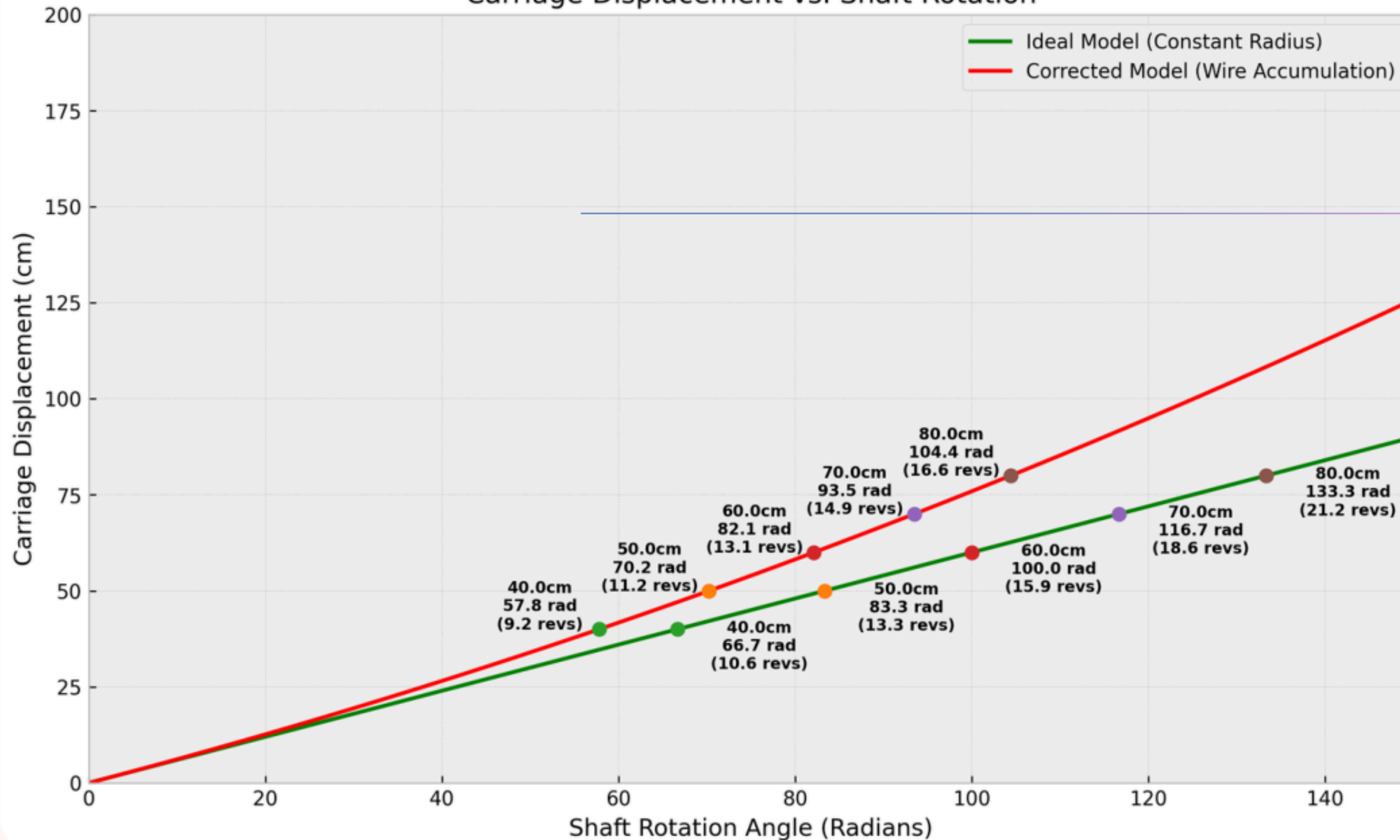
Structures & Materials:

Conduct check for any deflection of the rails (safety factor, failure and stress analysis)

Final evaluation of feasibility and market viability

CaDDE

Carriage Displacement vs. Shaft Rotation



Wire radius (**0.1mm**)

Idea Model:

Constant radius

Wire accumulation model:

Effective radius grows

$$r_{eff} = r_{shaft} + \left(r_{initial} \times \left(\frac{\theta}{\pi} \right) \right)$$

DC Motor chosen: 12V 18RPM w/ Encoder

No Load RPM: **18** ±10%

Rated Load Operating Current: 0.2A

Rated Load RPM: **13** ±10%

Rated Load Torque: 12kg (**117.7N**)

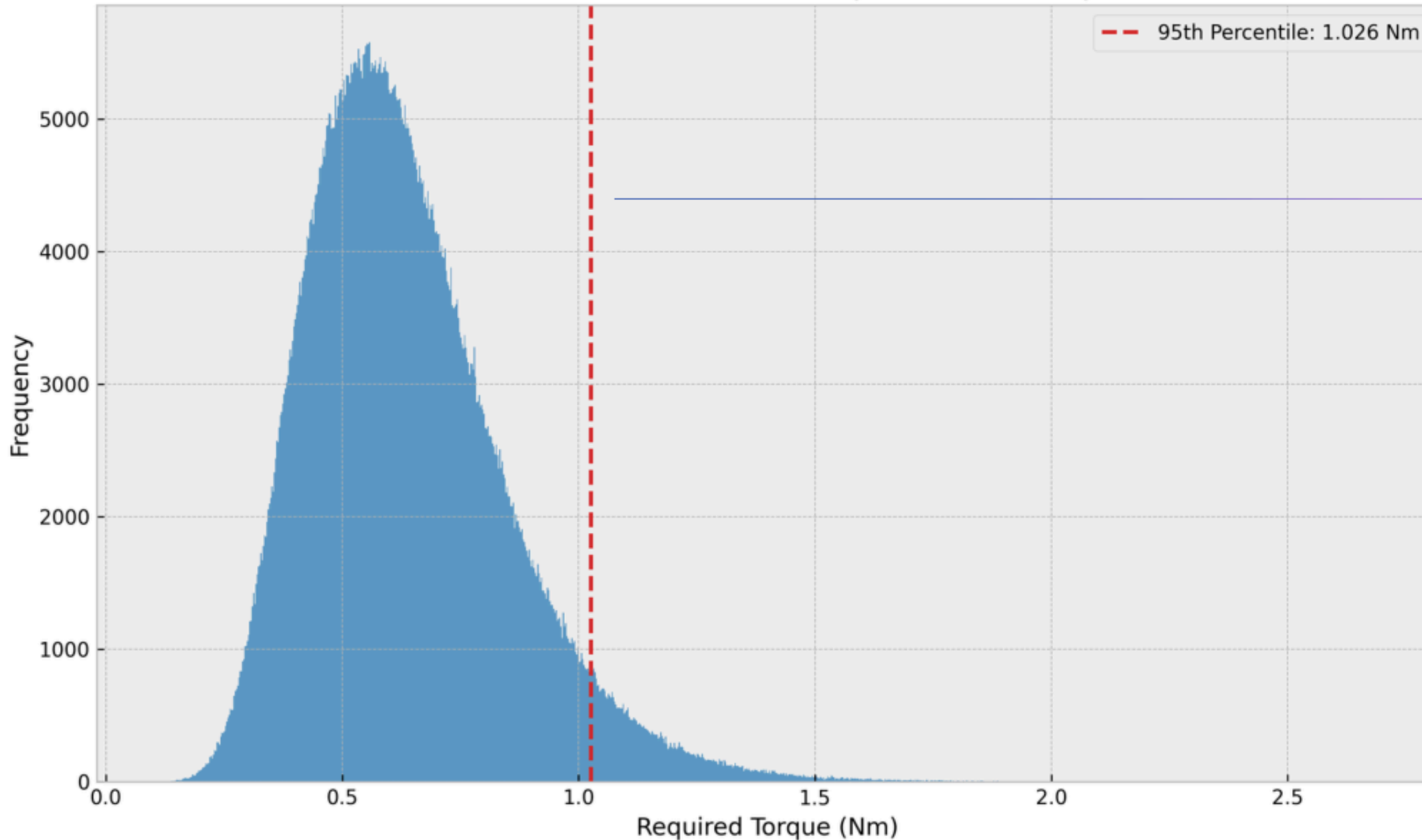
Rated Load Power: 2.4W

Stall Torque: 34kg (**333.5N**)

Stall Current: 1.3A

CaDDE

Monte Carlo: Distribution of Required Motor Torque



User mass: Normal distribution

(Mean: 60kg; Standard Deviation: 10kg)

Friction Coefficient: log normal distribution

(Mean: 0.18; Standard Deviation: 0.05)

Shaft Radius: Normal distribution

(Mean: 0.006m; Standard Deviation: 0.0003m)

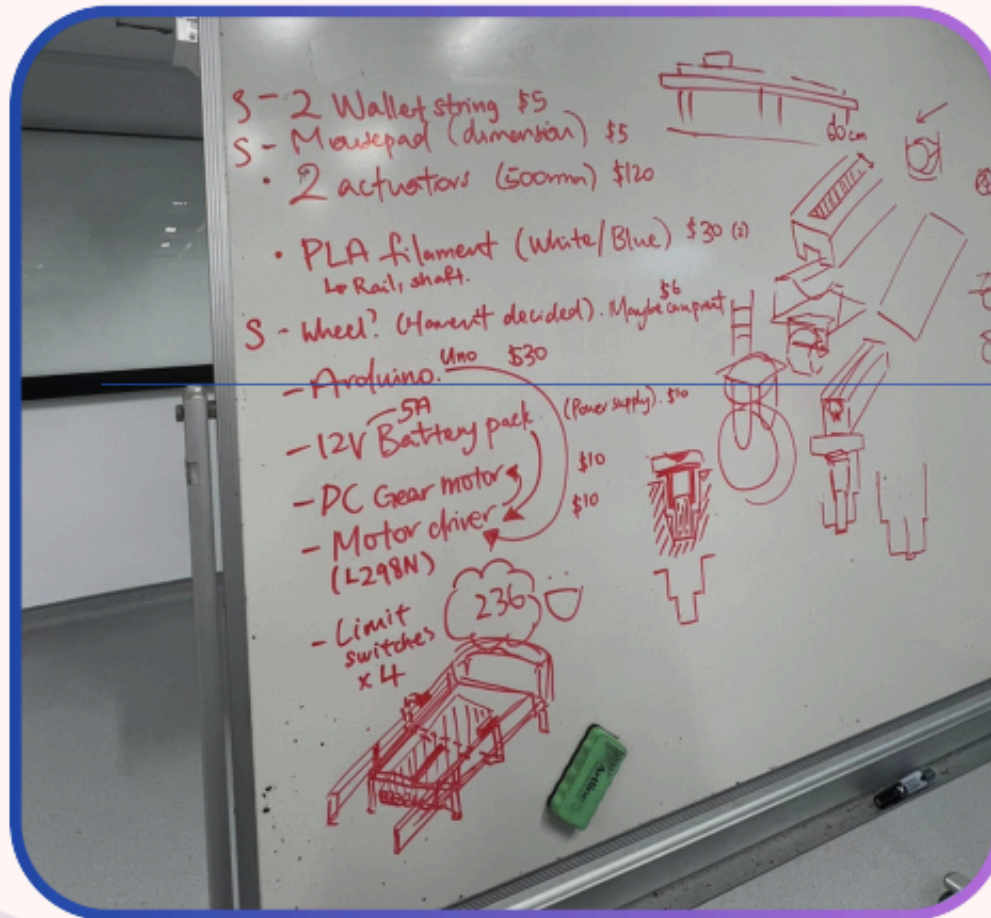
$$T = \mu \times m \times g \times r$$

Based on **1,000,000** simulations, we determined that our motor must provide at least **1.026 Nm** of torque to reliably assist **95%** of our target demographic.

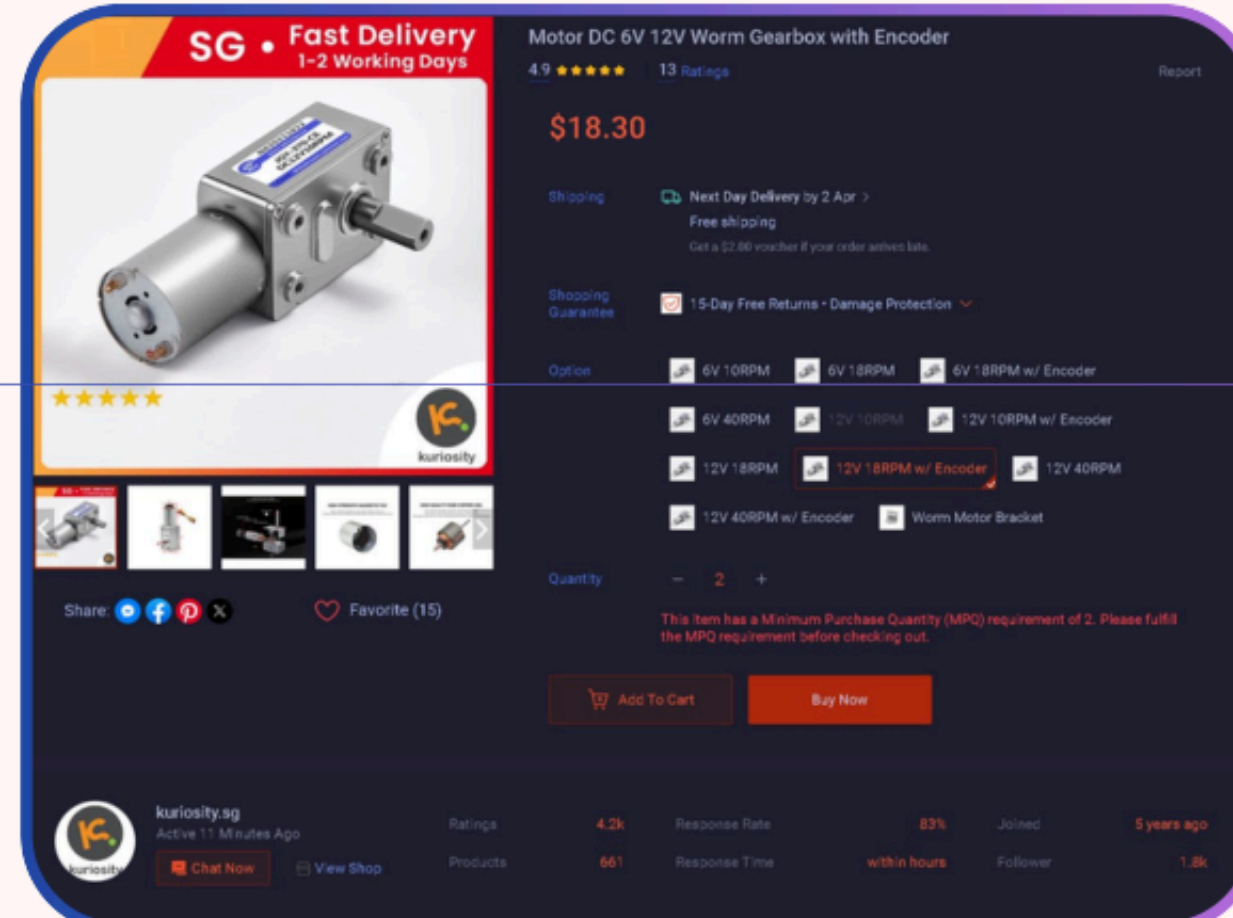
With a safety factor of 2, we require **2.052Nm total** motor torque.

Using a dual motor setup, each motor needs **1.026Nm** normal operating torque.

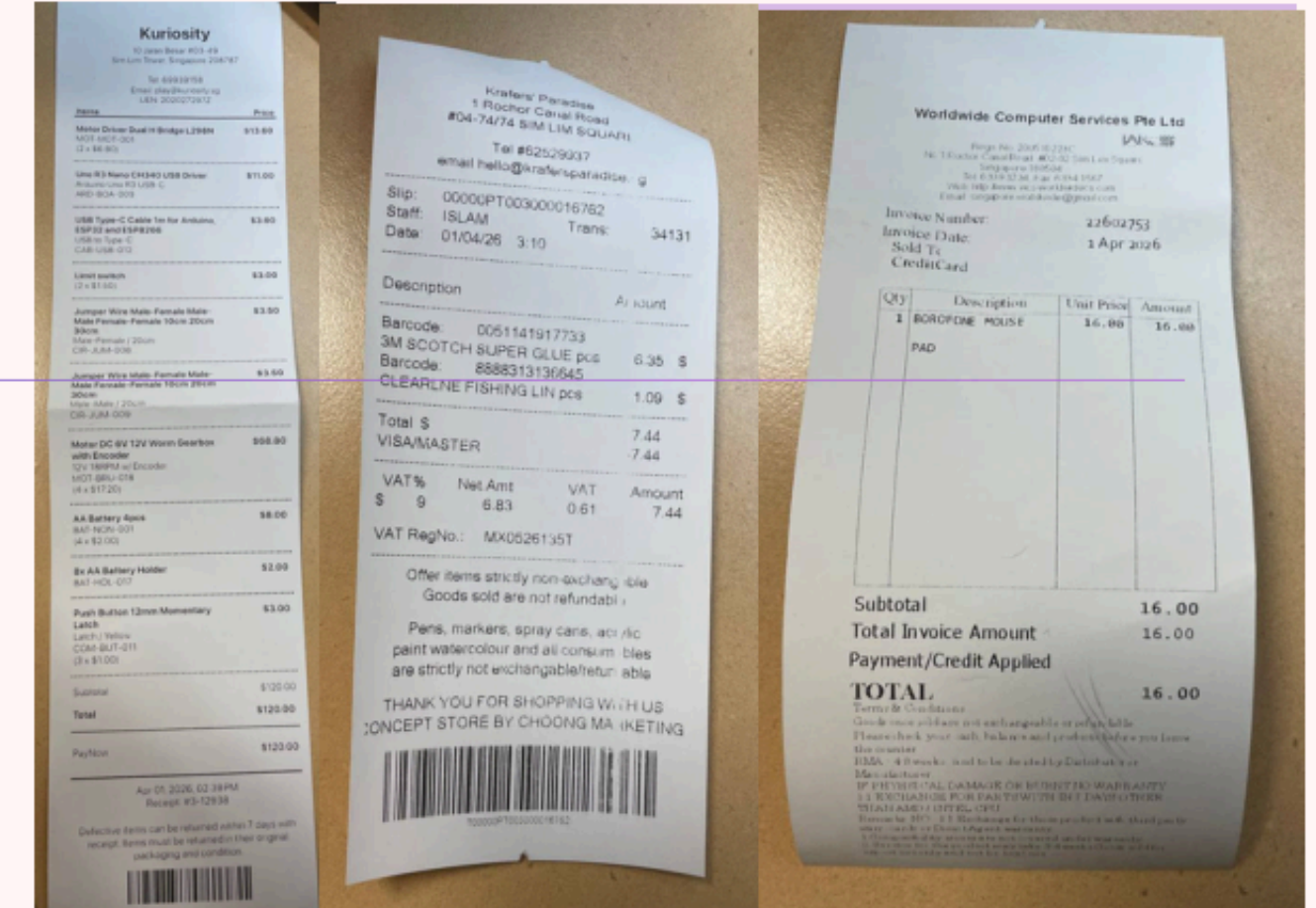
Purchasing Logistics



Confirming prototype,
brainstorming corresponding
logistics



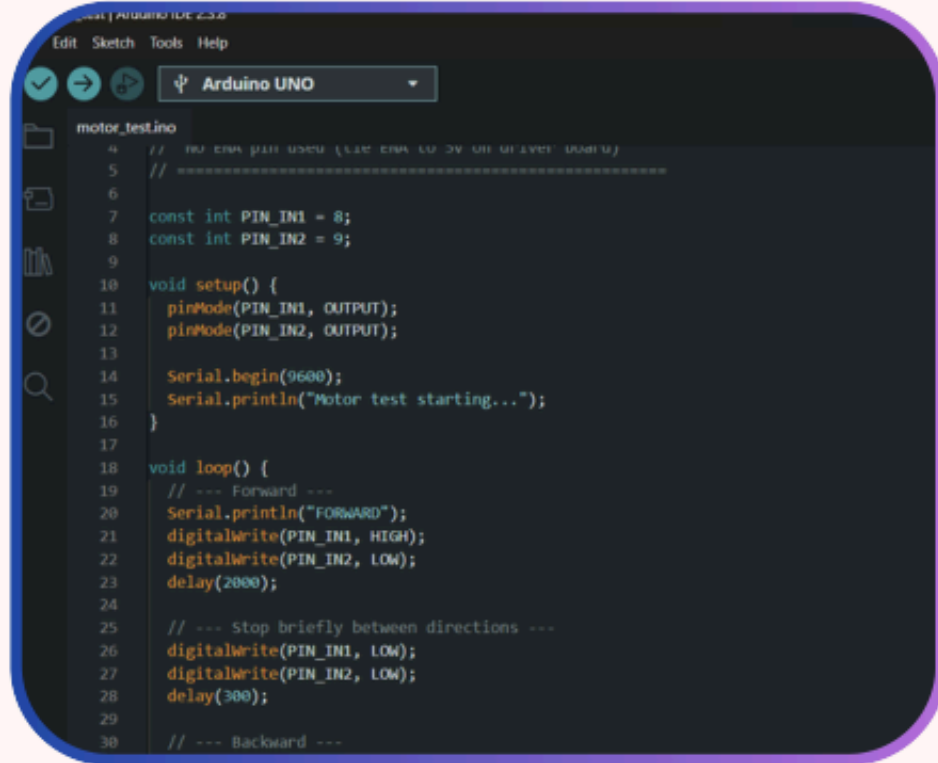
Confirming using online stores on what
models are available



Items purchased from Sim Lim and online!

Pending the arrival of the shaft

Circuit Connections



```
motor_test.ino
// Arduino IDE 2.3.0
Edit Sketch Tools Help

const int PIN_IN1 = 8;
const int PIN_IN2 = 9;

void setup() {
  pinMode(PIN_IN1, OUTPUT);
  pinMode(PIN_IN2, OUTPUT);

  Serial.begin(9600);
  Serial.println("Motor test starting...");
}

void loop() {
  // --- Forward ---
  Serial.println("FORWARD");
  digitalWrite(PIN_IN1, HIGH);
  digitalWrite(PIN_IN2, LOW);
  delay(2000);

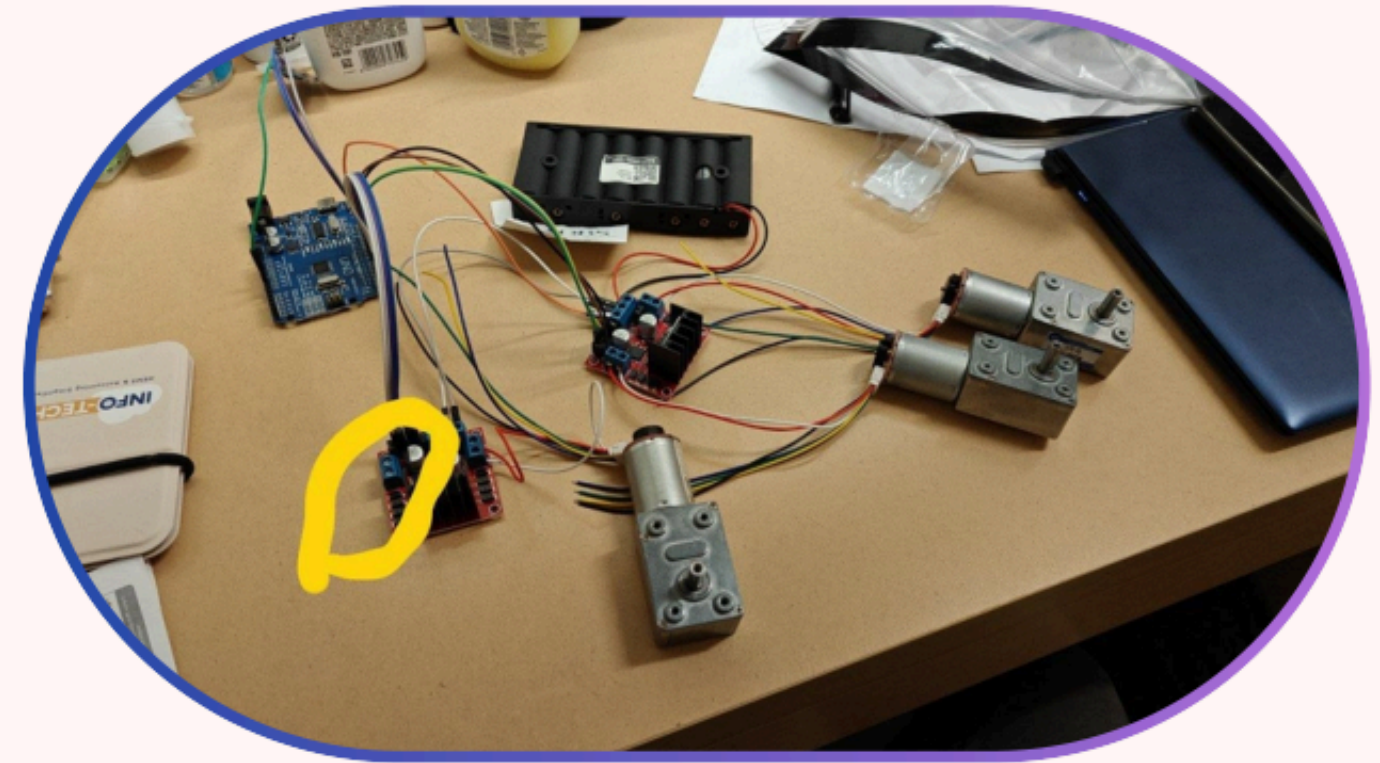
  // --- Stop briefly between directions ---
  digitalWrite(PIN_IN1, LOW);
  digitalWrite(PIN_IN2, LOW);
  delay(300);

  // --- Backward ---
}
```

Connecting Motor, Arduino and Motor driver first to testing motor



- Discovered 1 Motor Driver not working
- Purchased new Motor Driver



Testing the connections

Soldering connections together

Do up Arduino code and **test the code**

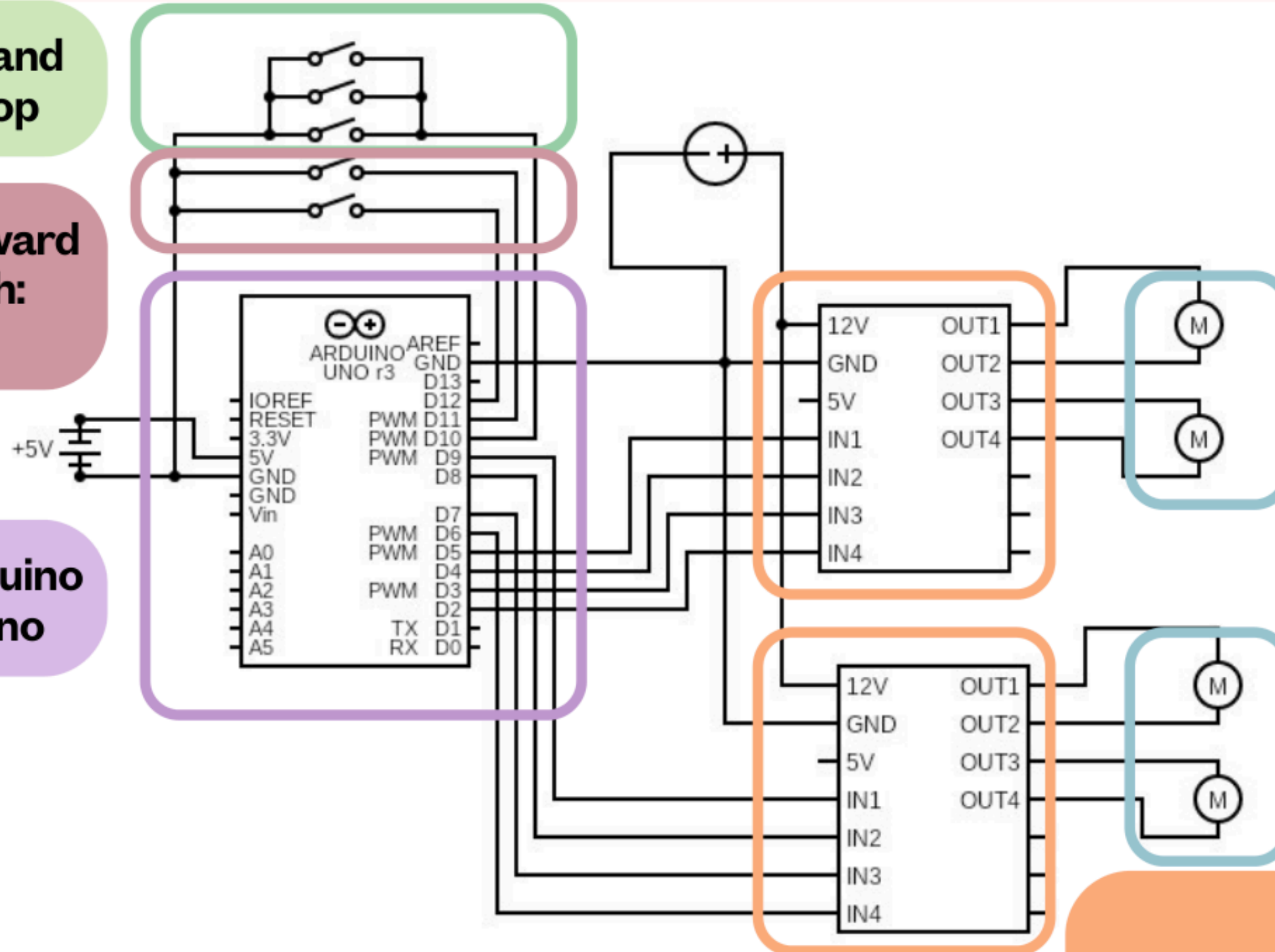
Circuit Connections

Connections

**Limit Switches and
Emergency stop**

**Top switch: Forward
Bottom switch:
Backward**

**Arduino
Uno**



Motors

- Set of 2 at the front and end of bed respectively

Motor Drivers

- To control speed, direction of motors etc.

Circuit Connections

INALCODE_3D.ino

```
1 // Motor pins
2 int motors[4][2] = {
3   {2, 3}, // Motor 1
4   {4, 5}, // Motor 2
5   {6, 7}, // Motor 3
6   {8, 9}  // Motor 4
7 };
8
9 // Switches
10 int fwSwitch = 10;
11 int revSwitch = 11;
12 int stopSwitch = 12;
```

The system takes user input through 3 switches:

- Forward
- Reverse
- Emergency Stop

We use INPUT_PULLUP, so:

- Not pressed = HIGH
- Pressed = LOW

```
void loop() {
28
29   // Emergency stop (highest priority)
30   if (digitalRead(stopSwitch) == LOW) {
31     stopAllMotors();
32     return;
33   }
34
35   // Forward
36   if (digitalRead(fwSwitch) == LOW) {
37     moveForward();
38   }
39
40   // Reverse
41   else if (digitalRead(revSwitch) == LOW) {
42     moveReverse();
43   }
44
45   // No button pressed → stop
46   else {
47     stopAllMotors();
48   }
49 }
```

Logic flow:

1. Emergency stop first (highest priority)
2. Then forward
3. Then reverse
4. Otherwise → stop

```
// ===== FUNCTIONS =====
52
53 void moveForward() {
54   for (int i = 0; i < 4; i++) {
55     digitalWrite(motors[i][0], HIGH);
56     digitalWrite(motors[i][1], LOW);
57   }
58 }
59
60 void moveReverse() {
61   for (int i = 0; i < 4; i++) {
62     digitalWrite(motors[i][0], LOW);
63     digitalWrite(motors[i][1], HIGH);
64   }
65 }
66
67 void stopAllMotors() {
68   for (int i = 0; i < 4; i++) {
69     digitalWrite(motors[i][0], LOW);
70     digitalWrite(motors[i][1], LOW);
71   }
72 }
```

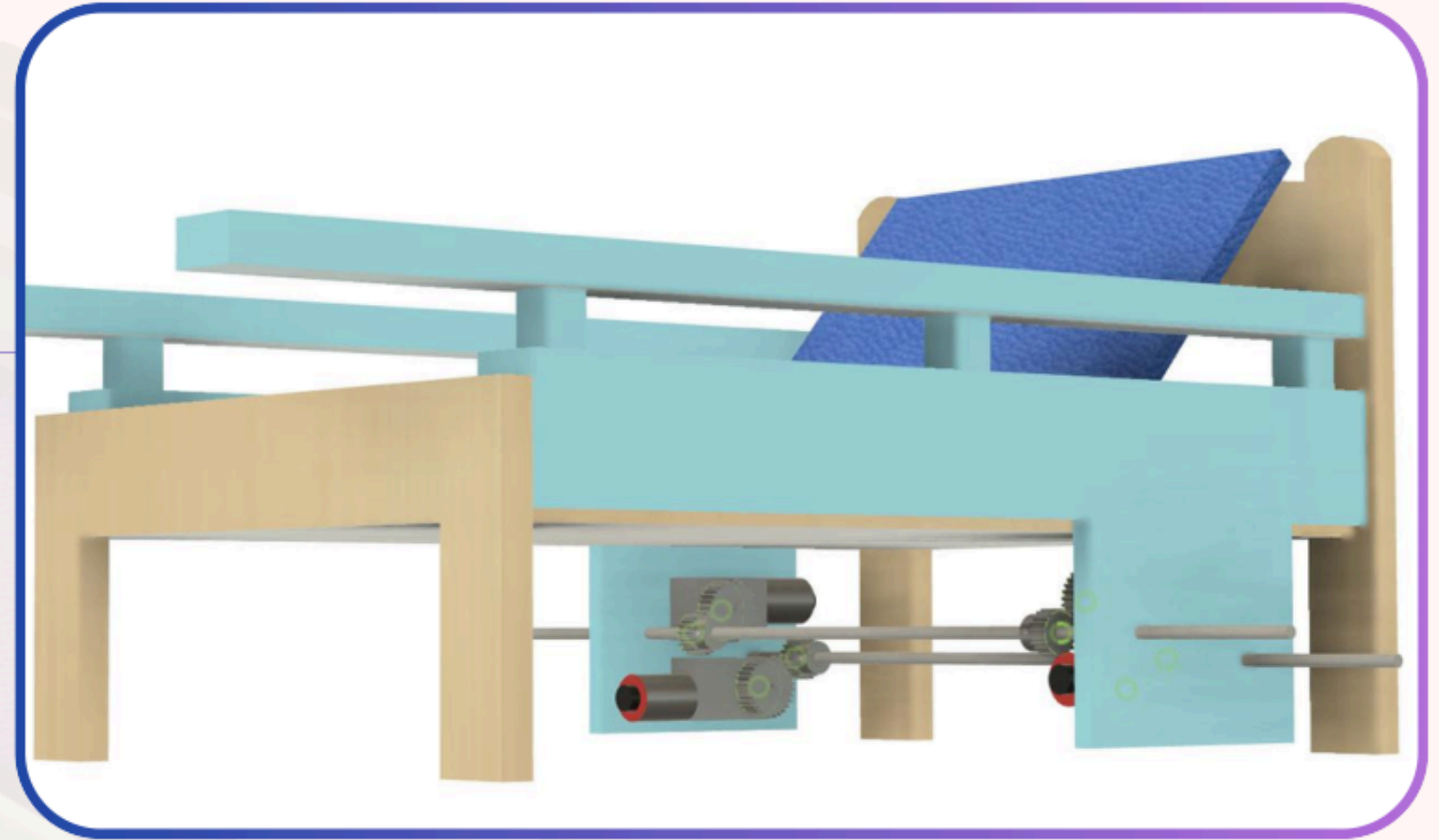
- Forward → HIGH / LOW
- Reverse → LOW / HIGH
- Stop → LOW / LOW

By changing the polarity, we control the motor direction.

Physical Assembly



Cutting the ends of the bed

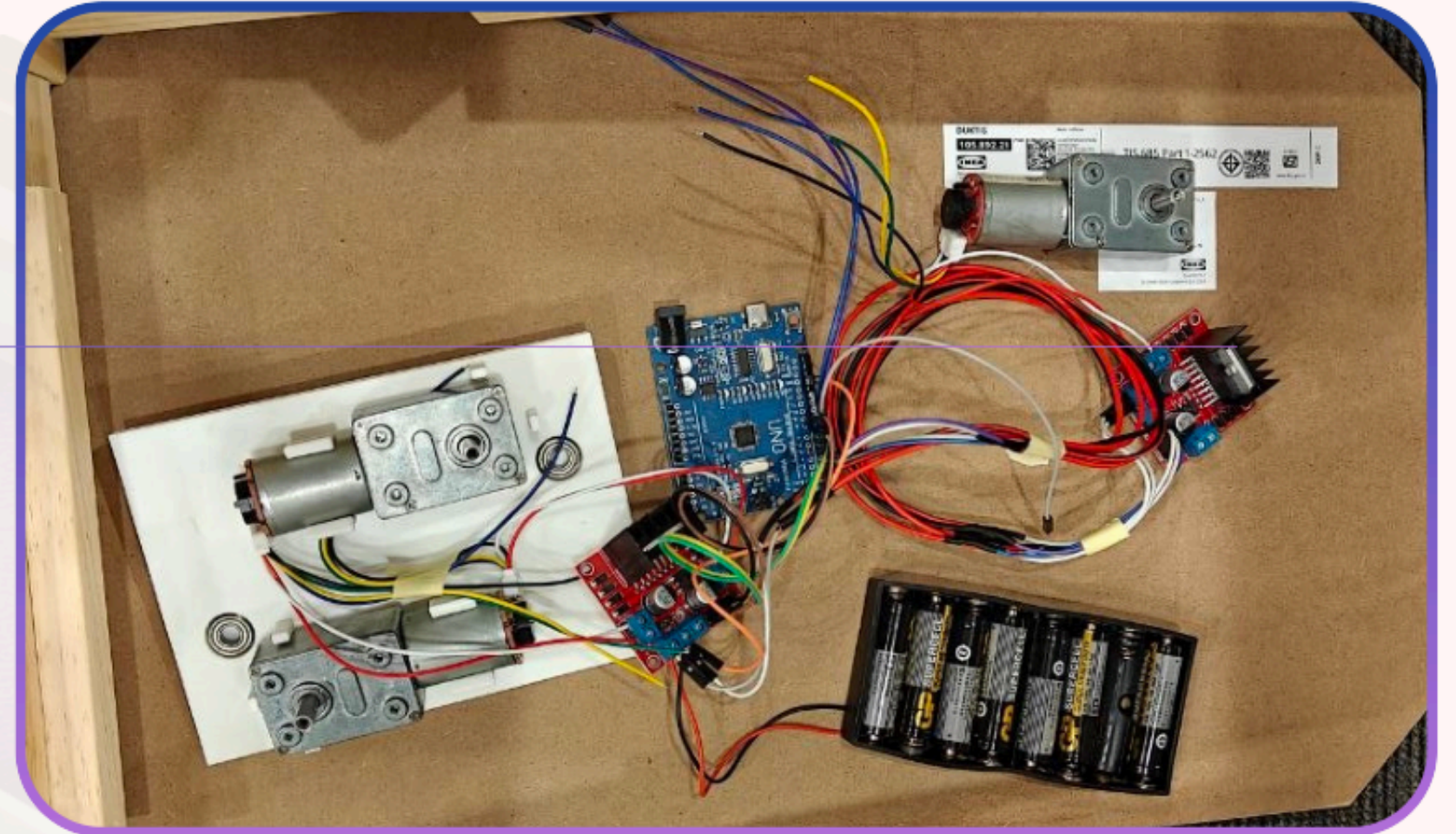


CAD out the rail to hold the shafts and motors, for 3D printing

Physical Assembly



“Conveyer” portion of the prototype



Actual Circuit



Challenges Faced

Unidirectional

Initial pulley system was one-way only, making it hard for the user to get back on.

Spatial Constraints

Planned on using linear actuators, but realised that the folded length of actuators would be too large and bulky

Ergonomic Integration

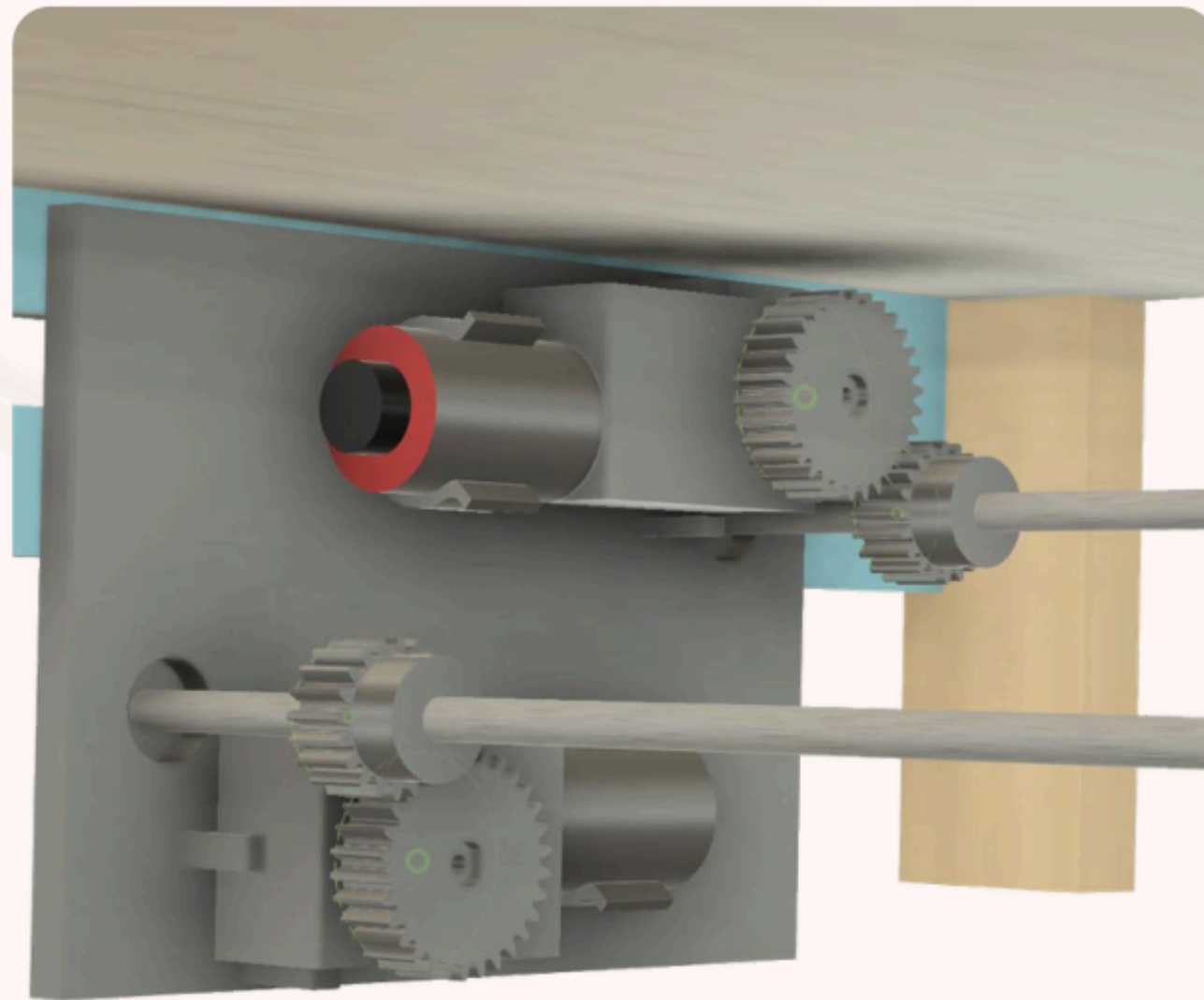
System had to be powerful to move a person, yet invisible and convenient enough that it doesn't feel like a scary piece of heavy machinery

Tackling Challenges

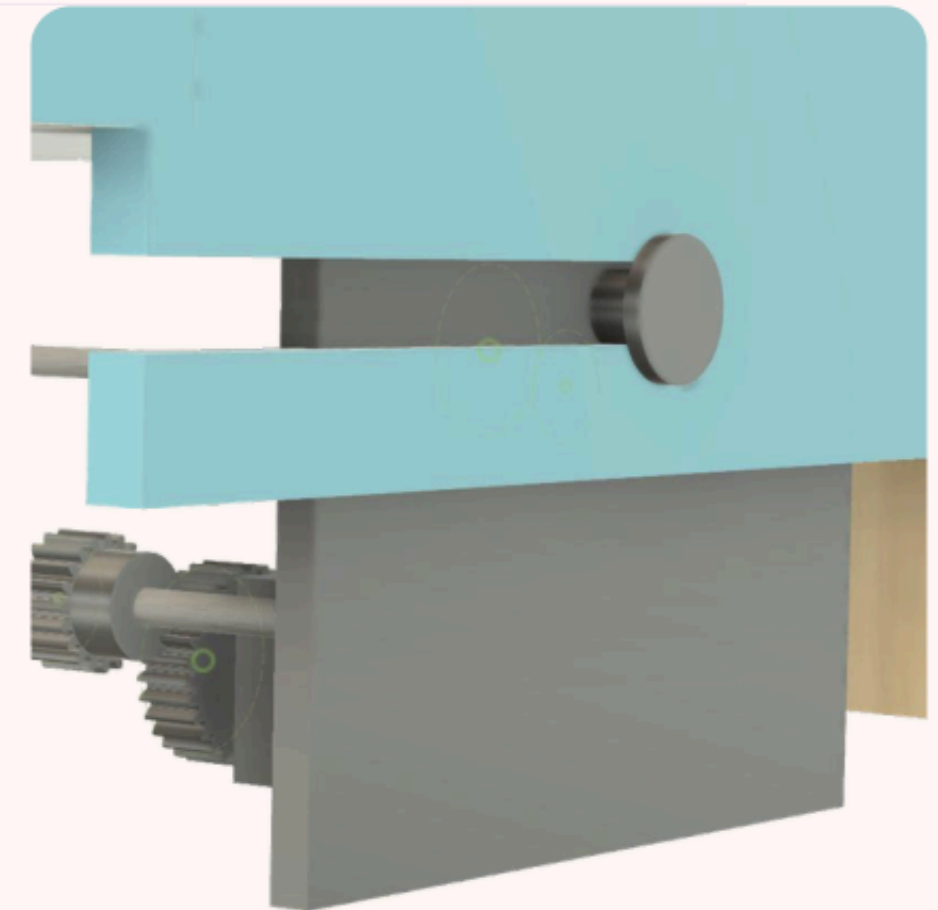
Bidirectional



**Synchronisation
of Motors**



**Detachable
Mechanism**

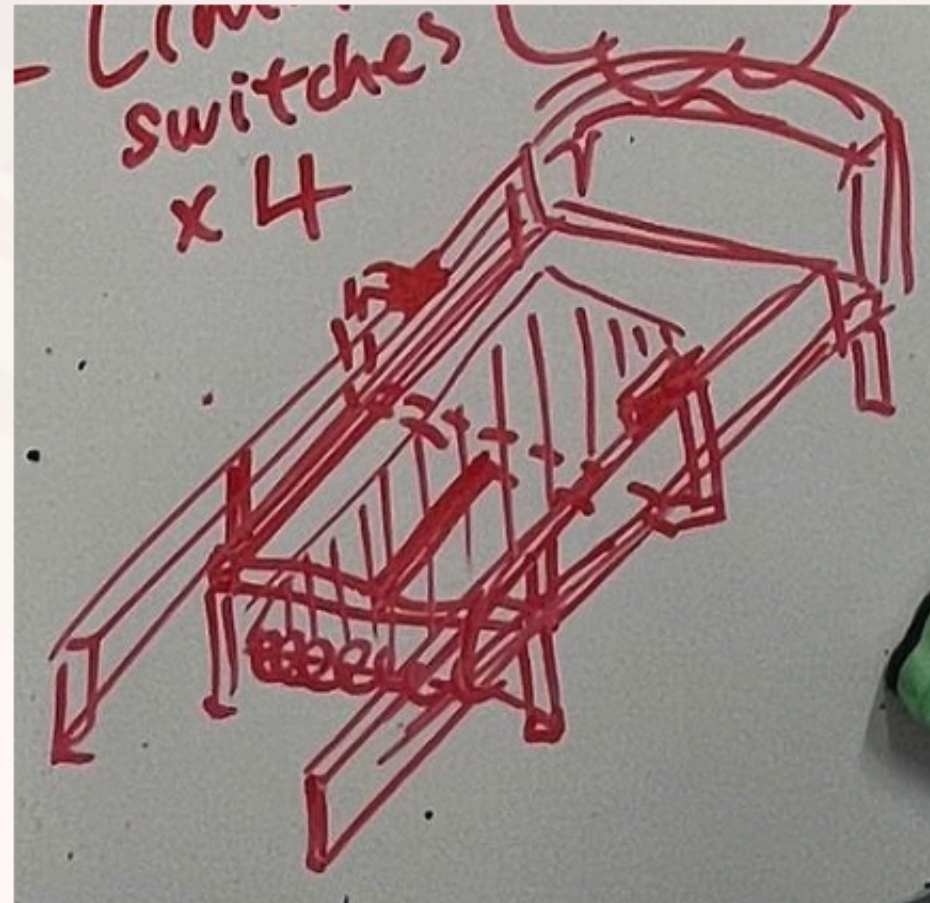


Iterations

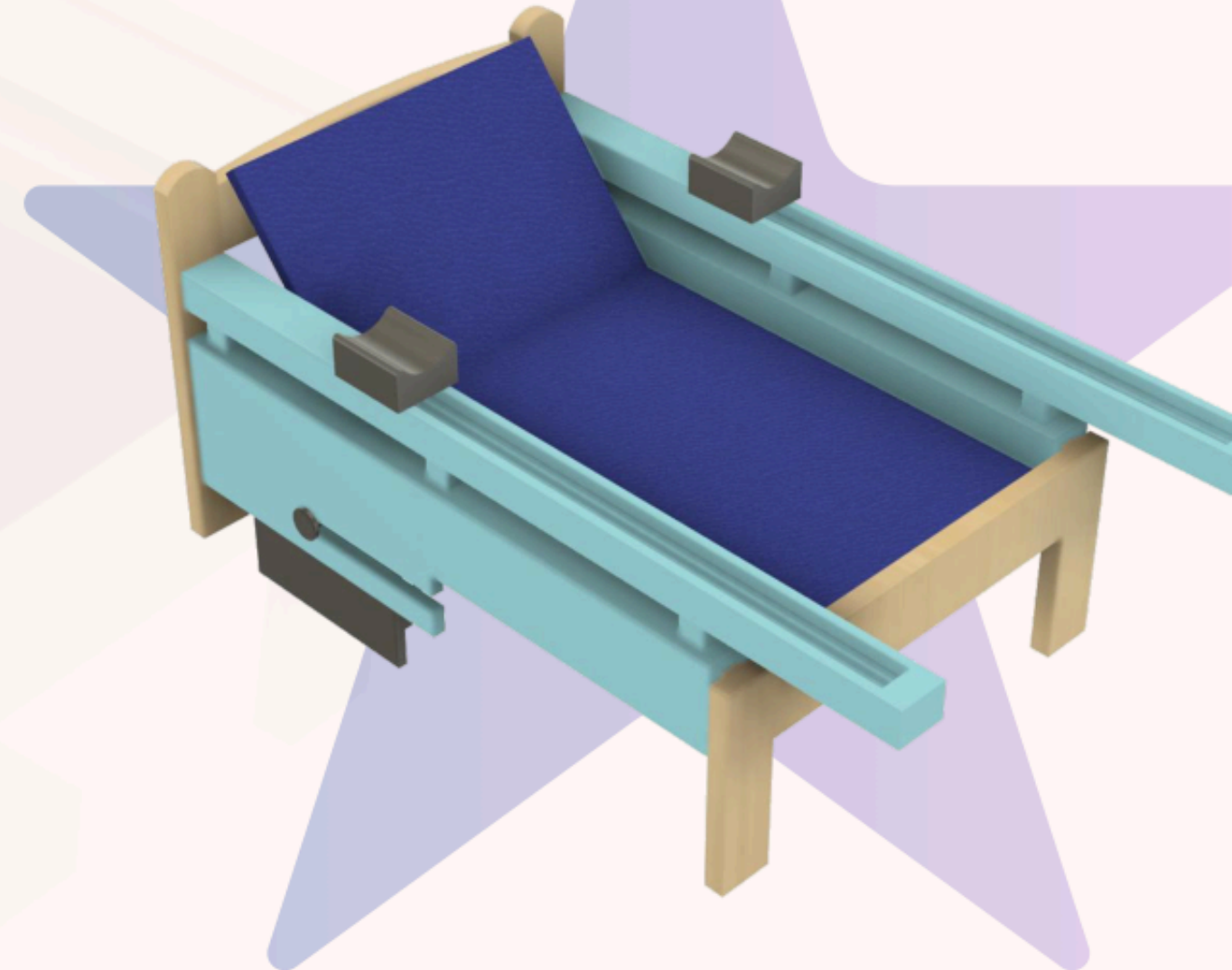
Pulley System



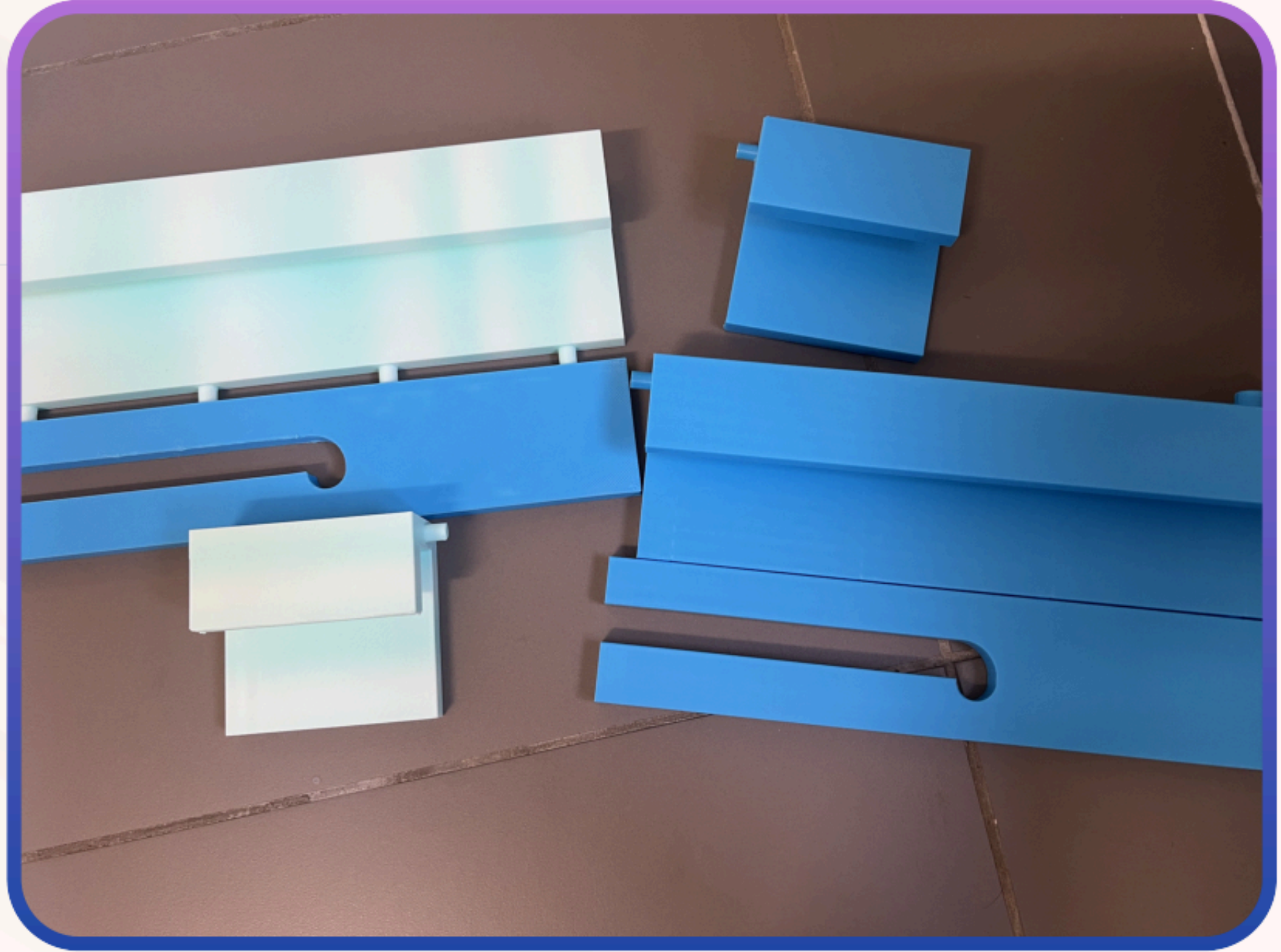
**Floating
Arms**



Final



Iterations



Thank you

